

- Solve all the problems below in **3 hours**.
- **Mobile phones are not allowed in the examination hall.**
- Please justify all your answers through proof or counterexample.
- Scoring: **1** $\rightarrow$ 2 points, **2** $\rightarrow$ 4 points, **3** $\rightarrow$ 2 points, **4** $\rightarrow$ 4 points, **5** $\rightarrow$ 2, **6** $\rightarrow$ 3, **7** $\rightarrow$ 3,.

1. Let f, g be two functions defined from $D \subset \mathbb{R}$ into $\mathbb{R}$. Translate using quantifiers the following statements (i.e: Write in mathematical language):

- 1) f is bounded; 2) f is odd; 3) f is never equal to 0; 4) f is periodic;
5) f is increasing; 6) f is not the 0 function; 7) f does not have the same value at two different points; 8) f is not less than g .

2. Solve in $\mathbb{R}$

$$\left\lfloor \frac{x+1}{2} \right\rfloor + \left\lfloor \frac{x+1}{4} \right\rfloor + \left\lfloor \frac{x+2}{4} \right\rfloor = [x] + 1.$$

3. (a) Solve in $\mathbb{C}$, the following equation:

$$x^2 + x + 1 = 0.$$

Let j be the solution with a positive imaginary part.

- (b) Show that for all $a, b \in \mathbb{R}$, $a + bj = 0 \implies a = b = 0$.

- (c) Let P be the polynomial function defined by

$$P(x) = x^7 + \alpha x^5 - x^4 + \beta x^3 - 2x^2 - 1, \text{ where } \alpha, \beta \in \mathbb{R}.$$

Determine α and β so that j is a solution of $P(x) = 0$.

4. Let (x_n) and (y_n) be the sequences defined for $n \in \mathbb{N}$ by

$$x_n = \sqrt{2n} - \sqrt{2n-1} \text{ and } y_n = \sqrt{n + (-1)^{n+1}} - \sqrt{n}.$$

Set

$$\Delta = \{x_n : n \in \mathbb{N}\} \text{ and } \Omega = \{y_n : n \in \mathbb{N}\}.$$

- (a) Show that Δ is bounded and determine $\max \Delta$, $\sup \Delta$, $\inf \Delta$ and $\min \Delta$ if they exist.

- (b) Find the relationship between Δ and Ω , then conclude for $\max \Omega$, $\sup \Omega$, $\inf \Omega$ and $\min \Omega$.

5. Prove that the sequence (U_n) defined by:

$$\begin{cases} U_0 = 0, \\ U_n = \frac{1}{2} \sin(U_{n-1}) + \frac{1}{3} \cos(U_{n-1}) \text{ for all } n \in \mathbb{N} \end{cases}$$

is convergent.

6. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be the function defined by $f(x) = \sin(x^2)$. Prove that:

- (a) f is uniformly continuous on any interval $[0, a]$ with $a > 0$,
(b) f isn't uniformly continuous on $[0, +\infty)$.

7. Let $F : (0, +\infty) \rightarrow (0, +\infty)$ be a continuous function satisfying

$$\lim_{x \rightarrow +\infty} \frac{F(x)}{x} = k > 1 > l = \lim_{x \rightarrow 0^+} \frac{F(x)}{x}.$$

Prove that the equation $F(x) = x$ has a solution.

Solution

1. : We have

$$1 \quad \exists M \in \mathbb{R}, \exists m \in \mathbb{R}, \forall x \in D : m \leq f(x) \leq M.$$

$$2 \quad \forall x \in D \implies -x \in D, \forall x \in D : f(-x) = -f(x).$$

$$3 \quad \forall x \in D : f(x) \neq 0.$$

$$4 \quad \exists a > 0, \forall x \in D \Rightarrow x + a \in D, f(x + a) = f(x).$$

$$5 \quad \forall (x, y) \in D^2, (x < y \implies f(x) < f(y)). \quad \mathbf{0.25 \text{ point for each}}$$

$$6 \quad \exists x \in D, f(x) \neq 0.$$

$$7 \quad \forall (x, y) \in D^2, (x \neq y \implies f(x) \neq f(y)).$$

$$8 \quad \exists x \in D, f(x) > g(x).$$

2. Set

$$Z_x = \left\lfloor \frac{x+1}{2} \right\rfloor + \left\lfloor \frac{x+1}{4} \right\rfloor + \left\lfloor \frac{x+2}{4} \right\rfloor.$$

For $x \in \mathbb{R}$ we distinguish four cases.

Case 1. $[x] = 4p$ with $p \in \mathbb{Z}$. Equivalently we have $4p \leq x < 4p + 1$. Therefore,

$$2p + \frac{1}{2} \leq \frac{x+1}{2} < 2p + 1, \quad p + \frac{1}{4} \leq \frac{x+1}{4} < p + \frac{1}{2}, \quad p + \frac{1}{2} \leq \frac{x+2}{4} < p + \frac{3}{4},$$

leading to

$$Z_x = 2p + p + p = 4p = [x] \neq [x] + 1 \quad \mathbf{1 \text{ point}}$$

Case 2. $[x] = 4p + 1$ with $p \in \mathbb{Z}$. Equivalently we have $4p + 1 \leq x < 4p + 2$. Therefore,

$$2p + 1 \leq \frac{x+1}{2} < 2p + \frac{3}{2}, \quad p + \frac{1}{2} \leq \frac{x+1}{4} < p + \frac{3}{4}, \quad p + \frac{3}{4} \leq \frac{x+2}{4} < p + 1,$$

leading to

$$Z_x = 2p + 1 + p + p = 4p + 1 = [x] \neq [x] + 1 \quad \mathbf{1 \text{ point}}$$

Case 3. $[x] = 4p + 2$ with $p \in \mathbb{Z}$. Equivalently we have $4p + 2 \leq x < 4p + 3$. Therefore,

$$2p + \frac{3}{2} \leq \frac{x+1}{2} < 2p + 2, \quad p + \frac{3}{4} \leq \frac{x+1}{4} < p + 1, \quad p + 1 \leq \frac{x+2}{4} < p + \frac{5}{4},$$

leading to

$$Z_x = 2p + 1 + p + p + 1 = 4p + 2 = [x] \neq [x] + 1 \quad \mathbf{1 \text{ point}}$$

Case 4. $[x] = 4p + 3$ with $p \in \mathbb{Z}$. Equivalently we have $4p + 3 \leq x < 4p + 4$. Therefore,

$$2p + 2 \leq \frac{x+1}{2} < 2p + \frac{5}{4}, \quad p + 1 \leq \frac{x+1}{4} < p + \frac{5}{4} \leq \frac{x+2}{4} < p + \frac{6}{4},$$

leading to

$$Z_x = 2p + 2 + p + 1 + p + 1 = 4p + 4 = [x] + 1. \quad \mathbf{0.5 \text{ point}}$$

Thus the set of solution of $Z_x = [x] + 1$ is $\cup_{p \in \mathbb{Z}} [4p + 3, 4p + 4[$. $\mathbf{0.5 \text{ point}}$

3. We have

(a) The solutions of the equation are: $j = -\frac{1}{2} + \frac{i\sqrt{3}}{2}, \bar{j} = -\frac{1}{2} - \frac{i\sqrt{3}}{2}$. $\mathbf{0.5 \text{ point}}$

(b) Let a, b be two real numbers, such that $a + bj = 0$, this leads to:

$$\begin{cases} a - \frac{b}{2} = 0, \\ \frac{b\sqrt{3}}{2} = 0, \end{cases}, \text{ the unique solution is } a = b = 0. \quad \mathbf{0.5 \text{ point}}$$

(c) Remark that $j^3 = 1$, then $P(j) = 0$, takes the form:

$$P(j) = j + \alpha j^2 - j + \beta - 2j^2 - 1 = 0,$$

Using the same technique as (b), we get $\alpha = 2, \beta = 1$. $\mathbf{1 \text{ point}}$

4. **1.** We have for all $n \in \mathbb{N}$,

$$0 < x_n = \sqrt{2n} - \sqrt{2n-1} = \frac{1}{\sqrt{2n} + \sqrt{2n-1}} \leq \frac{1}{\sqrt{2} + 1}. \quad \mathbf{0.5 \text{ point}}$$

This shows that Δ is bounded. Therefore $\sup \Delta$ and $\inf \Delta$ exist.

Since for all $n \in \mathbb{N}$, $x_n \leq \frac{1}{\sqrt{2}+1} = x_1$, we conclude that

$$\max \Delta = \sup \Delta = x_1 = \frac{1}{\sqrt{2} + 1}. \quad \mathbf{0.5 \text{ point}}$$

Now, let $lb(\Delta)$ be the set of all lower bounds of Δ , that is

$$lb(\Delta) = \{m \in \mathbb{R} : m \leq x_n \forall n \in \mathbb{N}\}.$$

Since $x_n > 0$ for all $n \in \mathbb{N}$, we conclude that $(-\infty, 0] \subset lb(\Delta)$. For any $m \in lb(\Delta)$, $m \leq x_n$ for all $n \in \mathbb{N}$ implies

$$m \leq \lim x_n = \lim \frac{1}{\sqrt{2n} + \sqrt{2n-1}} = 0. \quad \mathbf{1 \text{ point}}$$

Hence, we conclude that $lb(\Delta) \subset (-\infty, 0]$, then $lb(\Delta) = (-\infty, 0]$.

At the end we conclude that $\inf \Delta = \max(-\infty, 0] = 0$ and because $\Delta \cap lb(\Delta) = \Delta \cap (-\infty, 0] = \emptyset$, $\min \Delta$ does not exist. $\mathbf{0.5 \text{ point}}$

2. We have $\Omega = \Omega_{\text{even}} \cup \Omega_{\text{odd}}$ where

$$\begin{aligned}\Omega_{\text{even}} &= \{y_n : n \text{ even}\} = \{y_{2n} : n \in \mathbb{N}\} \text{ and} \\ \Omega_{\text{odd}} &= \{y_n : n \text{ odd}\} = \{y_{2n-1} : n \in \mathbb{N}\}.\end{aligned}$$

Since for all $n \in \mathbb{N}$ we have

$$y_{2n} = \sqrt{2n-1} - \sqrt{2n} = -(\sqrt{2n} - \sqrt{2n-1}) = -x_n$$

and

$$y_{2n-1} = \sqrt{2n} - \sqrt{2n-1} = x_n,$$

we conclude that $\Omega_{\text{even}} = -\Delta$, $\Omega_{\text{odd}} = \Delta$ and $\Omega = \Delta \cup (-\Delta)$. **1 point**

At the end, since $\Delta \subset (0, +\infty)$ and $(-\Delta) \subset (-\infty, 0)$ we obtain

$$\sup \Omega = \sup \Delta = \max \Delta = \max \Omega = \frac{1}{1 + \sqrt{2}} \quad \mathbf{0.25 \text{ point}}$$

and

$$\inf \Omega = \inf (-\Delta) = -\sup \Delta = -\max \Delta = \min (-\Delta) = \min \Omega = -\frac{1}{1 + \sqrt{2}}. \quad \mathbf{0.25 \text{ point}}$$

5. For all $n \in \mathbb{N}$ we have

$$\begin{aligned}|U_{n+1} - U_n| &= \left| \frac{1}{2} (\sin(U_n) - \sin(U_{n-1})) + \frac{1}{3} (\cos(U_n) - \cos(U_{n-1})) \right| \\ &\leq \frac{1}{2} (|\sin(U_n) - \sin(U_{n-1})|) + \frac{1}{3} |\cos(U_n) - \cos(U_{n-1})| \\ &\leq \frac{1}{2} (|U_n - U_{n-1}|) + \frac{1}{3} |U_n - U_{n-1}| \\ &= \frac{5}{6} (|U_n - U_{n-1}|). \quad \mathbf{1 \text{ point}}\end{aligned}$$

Since $\frac{5}{6} \in (0, 1)$, we conclude (U_n) is contractant and so, (U_n) is convergent. **1 point**

6. 1. for all $x, y \in [0, a]$ we have

$$\begin{aligned}|\sin(x^2) - \sin(y^2)| &\leq |x^2 - y^2| = |x + y| |x - y| \\ &\leq (|x| + |y|) |x - y| \\ &\leq 2a |x - y|. \quad \mathbf{1 \text{ point}}\end{aligned}$$

This shows that f is lipshitzian on $[0, a]$. Therefore, f is uniformly continuous on $[0, a]$.

2. Let for $n \in \mathbb{N}$, $x_n = \sqrt{n\pi + \frac{\pi}{2}}$ and $y_n = \sqrt{n\pi}$. Then for all $n \in \mathbb{N}$ we have

$$|x_n - y_n| = \sqrt{n\pi + \frac{\pi}{2}} - \sqrt{n\pi} = \frac{\pi}{2(\sqrt{n\pi + \frac{\pi}{2}} + \sqrt{n\pi})} \leq \frac{4}{4\sqrt{n\pi}} = \frac{1}{\sqrt{n\pi}}$$

and

$$\begin{aligned}
|\sin(x_n^2) - \sin(y_n^2)| &= 2 \left| \sin\left(\frac{x_n^2 - y_n^2}{2}\right) \right| \left| \cos\left(\frac{x_n^2 + y_n^2}{2}\right) \right| \\
&= 2 \sin\left(\frac{\pi}{4}\right) \left| \cos\left(n\pi + \frac{\pi}{4}\right) \right| \\
&= 2 \sin\left(\frac{\pi}{4}\right) \left| \cos(n\pi) \cos\left(\frac{\pi}{4}\right) - \sin(n\pi) \sin\left(\frac{\pi}{4}\right) \right| \\
&= 2 \left(\frac{\sqrt{2}}{2}\right) \left| (-1)^n \left(\frac{\sqrt{2}}{2}\right) \right| \\
&= 1. \quad \mathbf{2 \text{ points}}
\end{aligned}$$

Therefore, taking $\epsilon_0 = 1$, for any $\delta > 0$ we have $|x_{n_\delta} - y_{n_\delta}| \leq \delta$, where $n_\delta = E\left(\frac{1}{\pi\delta^2}\right) + 1$, and $|f(x_{n_\delta}) - f(y_{n_\delta})| = 1$. This shows that f is not uniformly continuous on $[0, +\infty)$.

7. Let $\varepsilon > 0$ be such that

$$(k - \varepsilon) > 1 > (l + \varepsilon).$$

One can choose $\varepsilon = \min\left(\frac{k-1}{4}, \frac{1-l}{4}\right)$, in this case we have

$$\begin{aligned}
1 &= \frac{3+1}{4} > \frac{3l+1}{4} = l + \frac{1-l}{4} \geq l + \varepsilon \text{ and} \\
1 &= \frac{3+1}{4} < \frac{3k+1}{4} = k - \frac{k-1}{4} \leq k - \varepsilon. \quad \mathbf{1 \text{ point}}
\end{aligned}$$

From the definition of the limit we deduce that there is $A > 0, B > 0$ such that

$$\begin{aligned}
F(x) &\geq (k - \varepsilon)x \text{ for all } x \geq B, \text{ and} \\
F(x) &\leq (l + \varepsilon)x \text{ for all } 0 < x \leq A.
\end{aligned}$$

Consider the function $G : [A, B] \rightarrow \mathbb{R}$ with $G(x) = F(x) - x$ for all $x \in [A, B]$. Clearly G is continuous and we have

$$\begin{aligned}
G(A) &= F(A) - A \leq (l + \varepsilon)A - A = ((l + \varepsilon) - 1)A < 0, \\
G(B) &= F(B) - B \geq (k - \varepsilon)B - B = ((k - \varepsilon) - 1)B > 0. \quad \mathbf{1 \text{ point}}
\end{aligned}$$

By Intermediate Value Theorem, there exists $c \in (A, B)$ such that $G(c) = 0$, that is $F(c) = c$. This shows that the equation $F(x) = x$ admits a solution. $\mathbf{1 \text{ point}}$